```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[10, 20], [30, 40]])
B = np.array([[5, 10], [15, 20]])

print("Matrix A:\n", A)
print("Matrix B:\n", B)
print("Addition:\n", A + B)
print("Subtraction:\n", A - B)
print("Multiplication:\n", A @ B)

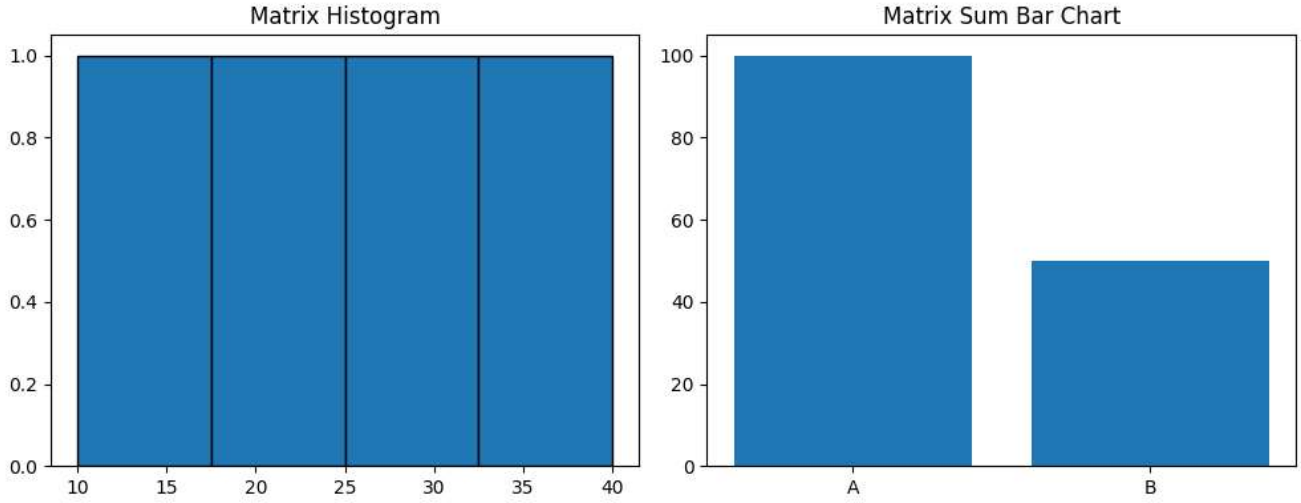
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.hist(A.flatten(), bins=4, edgecolor='black')
plt.title("Matrix Histogram")

plt.subplot(1, 2, 2)
plt.bar(["A", "B"], [A.sum(), B.sum()])
plt.title("Matrix Sum Bar Chart")

plt.tight_layout()
plt.show()
```

```
Matrix A:
 [[10 20]
 [30 40]]
Matrix B:
 [[ 5 10]
 [15 20]]
Addition:
 [[15 30]
 [45 60]]
Subtraction:
 [[ 5 10]
 [15 20]]
Multiplication:
 [[ 350  500]
 [ 750 1100]]
```




```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'Maths': [50, 60, 70, 80, 90],
    'Science': [55, 65, 75, 85, 95]
})

C = df.cov()
print("Covariance Matrix:\n", C)

values, vectors = np.linalg.eig(C.values)
print("\nEigenvalues:\n", values)
```

```
print("\nEigenvectors:\n", vectors)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.hist(df['Maths'], bins=5, edgecolor='black')
plt.title("Maths Marks Histogram")

plt.subplot(1, 2, 2)
plt.bar(["Maths", "Science"], values[:2])
plt.title("Eigenvalues Bar Chart")

plt.tight_layout()
plt.show()
```

```
Covariance Matrix:
          Maths  Science
Maths    250.0    250.0
Science  250.0    250.0

Eigenvalues:
 [500.   0.]

Eigenvectors:
 [[ 0.70710678 -0.70710678]
 [ 0.70710678  0.70710678]]
```

